

Steven A. Hauck, II

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Education

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| • Washington University in St. Louis | Earth and Planetary Sciences | Ph.D. – 2001 |
| • Washington University in St. Louis | Earth and Planetary Sciences | A.M. – 1998 |
| • University of Minnesota | Aerospace Engineering and Mechanics | B.A.E.M. – 1996 |

Appointments

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|------------------------------------|------------------------------------|----------------|
| • Department Chair | Case Western Reserve University | 2019 – Present |
| • Professor | Case Western Reserve University | 2015 – Present |
| • Associate Professor | Case Western Reserve University | 2009 – 2015 |
| • Assistant Professor | Case Western Reserve University | 2003 – 2009 |
| • Postdoctoral Associate | Carnegie Institution of Washington | 2001 – 2003 |
| • Graduate Research Assistant | Washington University in St. Louis | 1996 – 2001 |
| • Graduate Teaching Assistant | Washington University in St. Louis | 1996 – 1999 |
| • Undergraduate Teaching Assistant | University of Minnesota | 1995 – 1996 |
| • Undergraduate Research Assistant | University of Minnesota | 1994 |

Honors

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| • AGU Volunteer of the Month | 2024 |
| • Nominee, John S. Diekhoff Excellence in Graduate Teaching Award | 2023 |
| • Nominee, J. Bruce Jackson Award for Excellence in Undergraduate Mentoring | 2022 |
| • Interactive Commons Fellow, Case Western Reserve University | 2021 – 2022 |
| • NASA Group Achievement Award for MESSENGER Orbital Operations | 2017 |
| • UCITE Mentoring Fellow, Case Western Reserve University | 2013 – 2014 |
| • Nominee, John S. Diekhoff Excellence in Graduate Mentoring Award | 2011, 2015 |
| • MESSENGER Participating Scientist | 2007 – 2016 |
| • Carl Tolman Outstanding Graduate Teaching Assistant Award | 1998 – 1999 |
| ◦ <i>Department of Earth and Planetary Sciences, Washington University</i> | |
| • Letter of Recognition - Teaching Assistant | 1996 – 1997 |
| ◦ <i>Department of Earth and Planetary Sciences, Washington University</i> | |
| • Stephen E. Dwornik Planetary Geoscience Student Paper Award – Honorable Mention | 1997 |
| ◦ <i>Geological Society of America.</i> | |
| • National Merit Scholar | 1992 |

Selected Service to Community

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| • Council Member, AGU | 2024 – Present |
| • Council Leadership Team, AGU | 2025 – Present |
| • Chair, Publications Committee, AGU | 2024 – Present |
| • Member, Publications Committee, AGU | 2022 – Present |
| • Chair, Search Committee, Editor-in-Chief, <i>AGU Advances</i> | 2024 |
| • Search Advisory Committee, Executive Vice President of Science, AGU | 2023 – 2024 |

- Past-Chair, NASA Mercury Exploration Assessment Group (MExAG) 2023 – 2024
- Chair, NASA Mercury Exploration Assessment Group (MExAG) 2020 – 2023
- Search Committee, Editor-in-Chief, *JGR-Planets* 2023
- Search Committee, Editor-in-Chief, *JGR-Atmospheres* 2022
- Organizing Committee, MExAG 4th Annual Meeting (MExAG24) 2023–2024
- Convener, MExAG 3rd Annual Meeting (MExAG23) 2022–2023
- Convener, MExAG 2nd Annual Meeting (MExAG22) 2021–2022
- Convener, MExAG 1st Annual Meeting (MExAG21) 2020–2021
- Editor-in-Chief, *Journal of Geophysical Research – Planets* 2015 – 2019
- Steering Committee, AGU-AAS Promoting Interdisciplinary Research in Exoplanets (PIREX) Initiative 2018 – 2019
- Search Committee, Editor-in-Chief, *Earth and Space Science* 2018
- Session co-convener, 2018 AGU Fall Meeting, Joint AGU-AAS Frontiers in Exoplanets 2018
- AGU Digital Transformation Framework Working Group 2016
- Virtual Options Advisory Group, AGU 2015
- Organizer, 30th NASA MESSENGER Science Team Meeting, CWRU 2013
- Co-chair, Organizing Committee, *Workshop on the Evolution and Constitution of Mercury's Interior*, Chicago, IL 2012 – 2013
- Judge, Stephen E. Dwornik Planetary Geoscience Student Paper Award 2012
- National Research Council, Planetary Science Decadal Survey Inner Planets Panel 2009 – 2010
- Guest Editor, *Icarus*, MESSENGER Special Issue 2009 – 2010
- NASA Venus Exploration Analysis Group – *Planetary Formation and Evolution Focus Group* 2005 – 2009
- Associate Editor, *Journal of Geophysical Research – Planets* 2003 – 2007
- Session Organizer, Fall 2005 AGU Meeting, *Diversity of Planetary Magnetic Fields in the Solar System* 2005
- Member, NASA Mars Fundamental Research Program Review Panel 2003, 2004
- Member, NASA Mars Data Analysis Program Review Panel 2001, 2007
- Group Chief, NASA Planetary Geology and Geophysics Review Panel 2008
- Member, NASA Outer Planets Research Program Review Panel 2011
- Group Chief, NASA Planetary Mission Data Analysis Program Review Panel 2012
- Proposal Reviewer, NASA, NSF, Netherlands Organization for Scientific Research (NWO), Deutsche Forschungsgemeinschaft (German Funding agency DFG), Swiss National Science Foundation, Deutsches Zentrum für Luft- und Raumfahrt (German Aerospace Center, DLR), European Commission
- Peer Reviewer, *Astrobiology*, *Astrophysical Journal*, *Earth and Planetary Science Letters*, *Eos Transactions of the American Geophysical Union*, *Geochimica et Cosmochimica Acta*, *Geophysical Research Letters*, *Icarus*, *Journal of Geophysical Research*, *Nature*, *Physics of the Earth and Planetary Interiors*, *Planetary and Space Science*, *Progress in Earth and Planetary Science*, *Treatise on Geophysics*
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Institutional Service

- Artificial Intelligence Governance Committee 2025 – Present
- Artificial Intelligence Advisory Committee 2025
- Faculty Senate Executive Committee Subcommittee to Develop a Process for Creating, Splitting, Merging, Discontinuing, or Renaming Departments 2025 – Present
- Expanding Horizons Initiative Review Panel 2023

- Biology Department Chair Search Committee 2022
- Faculty Senate Committee on Information and Communication Technology (FSCICT) 2016 – 2022
- Chair, Faculty Senate Committee on Information and Communication Technology 2016 – 2020
- Advisory Committee on Research Computing 2005 – 2020
- Teaching and Learning Technology Advisory Committee 2020 – Present
- Coronavirus Technology and Pedagogy Faculty Leader 2020
- Faculty Senate 2016 – 2020
- Faculty Senate Committee on University Libraries 2016 – 2020
- VP of University Technology/CIO Cabinet 2015 – 2020
- Search Committee, Associate Vice President of University Technology 2017
- Faculty Sponsor for CWRU Information Technology Strategic Plan 2015 – 2016
- ITS Network Focus Group 2013 – 2014
- ITS Academic Technologies Faculty Support Site Focus Group 2012
- Admissions Event for Admitted Students and Alumni, Washington, DC 2012, 2013, 2017
- Faculty Senate Committee on Information and Communication Technology Fall 2012
- Freedman Fellows Selection Committee 2012, 2015
- Arts and Sciences Strategic Planning Task Force on Infrastructure 2008, 2014
- University Undergraduate Faculty Curriculum Committee 2005 – 2007
- Case High Performance Computing Technical Working Group 2004 – 2005

Department Service

- Departmental Representative & Undergraduate Advisor for EEPS Majors 2014 – Present
- Graduate Admissions Committee 2003 – Present
- Website Committee 2006 – Present
- Chair, Third-year Review Committee 2026
- Co-Chair, Instructor Faculty Search Committee 2025
- Chair, Promotion Committee 2024
- Chair, Departmental Review Committee 2023 – 2025
- Climate Systems Faculty Search Committee 2022 – 2023
- Journal Club Coordinator 2003 – 2020
- Chair, Visiting Faculty Search Committee 2021
- Visiting Faculty Search Committee 2018, 2019
- Chair, Special Promotion Committee 2017
- Ph.D. Program Review Self-Study Committee 2013 – 2014
- Faculty Search Committee 2012 – 2013
- Faculty Search Committee 2009 – 2010
- Department *Experience Case* Coordinator 2006 – 2009
- Faculty Search Committee 2004 – 2005
- Department Seminar Coordinator 2004 – 2006

Advising and Mentoring

- Post-doctoral: Nathalie C. Michel (2011 – 2013), Audrey Martin (w/J. Van Orman, 2013 – 2015), Ludovic Huguet (2016 – 2018)
- Ph.D: J. Andreas Ritzer (2004 – 2010), Katherine Crispin (w/J. Van Orman, 2005 – 2011), Jeffrey Balcerski (2008 – 2015), Sara Gutierrez (2018 – 2025).
- M.S. Robert Moore (w/J. Van Orman, 2013 – 2021)

- Undergraduate research: Derek Smith (Summer 2005), David Blair (2005–2007), Zachary Newman (2006–2007), Andrew Opsitnick (2007–2008), Andrew Enns (2008), Marinda Mitchell (Summer 2009–Spring 2010), Peng (Victoria) Sun (2011–2015), Rebecca Steely (Fall 2011), Mark Richardson (2011–2012), Emilie Dunham (2012–2013), Zachary Williams (2013–2014), Nihar Chhatiawala (2013–2014), Anthony Dario (2013–2014), Cullin Brown (Spring 2014), Brittany Adams (North Carolina State University, Summer 2015), Patrick Shober (2015–2017), Justo Karell (Fall 2015), Hannah Goldberg (Summer 2016), Heather Pantell (Astronomy Senior Thesis, 2016–2017), Megan Masterson (2017–2019), Elizabeth Heiny (2019–2021), Mariana Blanco (2021–2023, incl Physics Senior Thesis), Katelyn Lamm (2022–2023).
- High school research: Michael Butler (Hawken, 2018)
- Additional thesis advising/examination committees beyond principal mentees: Molly Yunker (M.S. 2005), Yan Liu (Ph. D. Civ. Eng. 2011), Bin (Benjamin) Zhang (Ph.D. student in Civ. Eng., 2007), Ruth Jacob (Ph.D. 2015), Alexander Stark (Ph.D., TU Berlin, Berlin, Germany, 2015), Jian Han (Ph.D. 2016), Julie Bloxson (Ph.D. 2017), Aaron Watkins (Ph.D. Astronomy, 2017), Zoe Zeszut (M.S. 2017), Jurrien Knibbe (Ph.D., Vrije Universiteit Amsterdam, Netherlands, 2018), Man Xu (Ph.D., 2020), Kristina Collins (Ph.D. in EECS, 2022), Ray Garner (Ph.D. Astronomy, 2023), Wesley Tucker (Ph.D., University of Illinois – Chicago, 2023), Jemila Edmond (Ph.D., 2024), Hope Omodolor (Ph.D., 2025), Zhihua Xiong (Ph.D., 2025)
- Case Western Reserve University Geological Society (CWRUGS) Advisor 2005 – 2010
- CWRU Students for the Exploration and Development of Space (SEDS) Advisor 2020 – Present

Professional Society Memberships

- American Geophysical Union 1994 – Present
- Geological Society of America 2011 – Present
- National Association of Geoscience Teachers 2017 – Present

Invited Professional Presentations

- AGU – Fall Meeting, Building Policy for Wide-Open Science Panel 2024
- Planetary Science Technology Symposium 2024
- NASA Planetary Science Advisory Committee March and June 2023
- NASA Planetary Science Advisory Committee February and December 2022
- NASA Planetary Science Advisory Committee March, June, and November 2021
- NASA Planetary Science Advisory Committee August and November 2020
- National Research Council, Mercury and the Moon Panel, Planetary Science and Astrobiology 2023–2032 Decadal Survey 2020
- Study of the Earth’s Deep Interior 2018
- Mercury: Current and Future Science of the Innermost Planet 2018
- Western Michigan University 2017
- Hyland Software, Cleveland 2015
- BepiColombo – MESSENGER Joint Meeting, Berlin, Germany 2015
- Case Western Reserve University – Department of Astronomy 2015
- Carnegie Institution for Science – Geophysical Laboratory 2014
- Case Western Reserve University – Department of Physics 2014
- Ohio Section of the American Association of Physics Teachers 2013
- Gordon Research Conference – Interior of the Earth 2013
- AGU Fall Meeting – Terrestrial Planetary Cores Session 2011
- Case Western Reserve University – Department of Geological Sciences 2010
- University of Akron 2010
- Southwest Research Institute, Boulder 2010
- University of California – Los Angeles 2007

- JASON Project Professional Development Workshop 2005
- Case Western Reserve University – Department of Astronomy 2004
- University of Illinois at Chicago 2003
- Case Western Reserve University 2003
- 34th Interface Foundation Symposia, Montreal 2002
- Princeton University 2001
- Brown University – Vernadsky Institute Microsymposium 1999

Invited Public Lectures

- Science Café Cleveland (w/ J. Van Orman) 2018
- Shaker Heights Libraries Science Café 2018
- AGU Ask-Me-Anything on Reddit 2016
- College Club of Cleveland 2015
- Pluto Flyby Event, Institute for Science of Origins and Happy Dog 2015
- Science Café Cleveland 2015
- Akron Physics Club 2015
- Frontiers of Astronomy Lecture Series, CWRU Dept. of Astronomy, Cleveland Astronomical Society, and Cleveland Museum of Natural History 2013
- Cleveland Astronomical Society 2013
- Baldwin-Wallace University, Burrell Memorial Observatory 2012
- Akron Physics Club 2011
- Cleveland Astronomical Society 2010
- Third Saturday Series, NASA Glenn Research Center Visitor's Center 2008
- Frontiers of Astronomy Lecture Series, CWRU Dept. of Astronomy, Cleveland Astronomical Society, and Cleveland Museum of Natural History 2005
- Cleveland Astronomical Society 2005
- Ecophilia 2004
- Northern Ohio Geological Society 2004

Workshops

- Author and Peer Reviewer Workshop, Co-leader, LPSC, Houston, TX 2018
- Workshop on Venus Exploration Targets, Houston, TX 2014
- Workshop on the Evolution and Constitution of Mercury's Interior, Chicago, IL 2013
- Early Career Faculty Workshop: Teaching, Research, and Managing Your Career, College of William and Mary 2004

Publication in peer-reviewed journals and books (Hauck, his students and post-docs in bold, § = student author)

- **§Moore, R. D.**, J. A. Van Orman, **S. A. Hauck, II**, The influence of oxygen and sulfur on the partitioning of lithophile elements into liquid iron alloys, *Geochimica et Cosmochimica Acta*, in revision.
- Ernst, C. M., N. L. Chabot, R. L. Klima, S. Kubota, G. Rogers, P. K. Byrne, **S. A. Hauck, II**, K. E. Vander Kaaden, R. L. Vervack, Jr., S. Besse, D. T. Blewett, B. W. Denevi, S. Goosens, S. J. Indyk, N. R. Izenberg, C. L. Johnson, L. M. Jozwiak, H. Korth, R. L. McNutt, Jr., S. L. Murchie, P. N. Peplowski, J. M. Raines, E. B. Rampe, M. S. Thompson, S. Z. Weider, Science goals and mission concept for a landed investigation of Mercury, *Planetary Science Journal*, doi: 10.3847/PSJ/ac1c0f, 2022.

- Genova, A., S. Goossens, E. Mazarico, F. G. Lemoine, G. A. Neumann, W. Kuang, T. Sabaka, **S. A. Hauck, II**, D. E. Smith, S. C. Solomon, M. T. Zuber, Geodetic evidence that Mercury has a solid inner core, *Geophysical Research Letters*, 467, 3625–3633, doi: 10.1029/2018GL081135, 2019.
- **Hauck, S. A., II**, and C. L. Johnson, Mercury: Inside the Iron Planet, *Elements*, 15, 21–26, doi: 10.2138/gselements.15.1.21, 2019.
- **Hauck, S. A., II**, M. Grott, P. K. Byrne, B. W. Denevi, S. Stanley, T. J. McCoy, Mercury's global evolution, in *Mercury after the MESSENGER Mission*, (S. C. Solomon, B. J. Anderson, and L. R. Nittler, Eds), Cambridge University Press, pp. 516-543, doi:10.1017/9781316650684.020, 2018.
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- **Huguet, L.**, J. A. Van Orman, **S. A. Hauck, II**, M. Willard, Earth's inner core nucleation paradox, *Earth and Planetary Science Letters*, 487, 9-20, doi: 10.1016/j.epsl.2018.01.018, 2018.
- **Hauck, S. A., II**, D. Baratoux, S. Stanley, Twenty five years of planetary science: Discoveries and new questions, *Journal of Geophysical Research: Planets*, 121, 1829-1830, doi: 10.1002/2016JE005185, 2016.
- Johnson, C. L., and **S. A. Hauck, II**, A whole new Mercury: MESSENGER reveals a dynamic planet at the last frontier of the inner solar system, *Journal of Geophysical Research: Planets*, 121, 2349-2362, doi: 10.1002/2016JE005150, 2016.
- Johnson, C. L., L. C. Philpott, B. A. Anderson, H. Korth, **S. A. Hauck, II**, D. Heyner, R. J. Phillips, R. M. Winslow, S. C. Solomon, MESSENGER observations of induced magnetic fields in Mercury's core, *Geophysical Research Letters*, 43, doi:10.1002/2015GL067370, 2016.
- Peale, S. J., J.-L. Margot, **S. A. Hauck, II**, S. C. Solomon, Consequences of a solid inner core on Mercury's spin configuration, *Icarus*, 264, 443-455, doi:10.1016/j.icarus.2015.09.024, 2016.
- Perry, M. E., G. A. Neumann, R. J. Phillips, O. S. Barnouin, C. M. Ernst, D. S. Kahan, S. C. Solomon, M. T. Zuber, D. E. Smith, **S. A. Hauck, II**, S. J. Peale, J.-L. Margot, E. Mazarico, C. L. Johnson, R. W. Gaskell, J. H. Roberts, R. L. McNutt, Jr., J. Oberst, The low-degree shape of Mercury, *Geophysical Research Letters*, 42, 6951-6958, doi:10.1002/2015GL065101, 2015.
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- Watters, T. R., M. M. Selvens, M. E. Banks, **S. A. Hauck, II**, K. J. Becker, M. S. Robinson, Distribution of large-scale contractional tectonic landforms on Mercury: Implications for the origin of global stresses, *Geophysical Research Letters*, 42, 3755-3763, doi: 10.1002/2015GL063570, 2015.
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- Peale, S. J., J.-L. Margot, **S. A. Hauck, II**, and S. C. Solomon, Effect of core-mantle and tidal torques on Mercury's spin axis orientation, *Icarus*, 231, 206-220 doi:10.1016/j.icarus.2013.12.007, 2014.
- **Hauck, S. A., II**, J.-L. Margot, S. C. Solomon, R. J. Phillips, C. L. Johnson, F. G. Lemoine, E. Mazarico, T. J. McCoy, S. Padovan, S. J. Peale, M. E. Perry, D. E. Smith, M. T. Zuber, The curious case of Mercury's internal structure, *Journal of Geophysical Research: Planets*, 118, 1204-1220, doi:10.1002/jgre.20091, 2013.
- **Michel, N. C., S. A. Hauck, II**, S. C. Solomon, R. J. Phillips, J. H. Roberts, M. T. Zuber, Thermal evolution of Mercury as constrained by MESSENGER observations, *Journal of Geophysical Research*, 118, 1033-1044, doi:10.1002/jgre.20049, 2013.
- Zolotov, M. Yu., A. L. Sprague, **S. A. Hauck, II**, L. R. Nittler, S. C. Solomon, S. Z. Weider, The redox state, FeO content, and origin of sulfur-rich magmas on Mercury, *Journal of Geophysical Research*, 118, doi:10.1029/2012JE004274, 2013.
- Dombard, A. J., **S. A. Hauck, II**, **§J. A. Balcerski**, On the origin of mascon basins on the Moon (and beyond), *Geophysical Research Letters*, 40, 28-32, doi:10.1029/2012GL054310, 2013.
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- Zuber, M. T., D. E. Smith, R. J. Phillips, S. C. Solomon, G. A. Neumann, **S. A. Hauck, II**, S. J. Peale, O. S. Barnouin, J. W. Head, C. L. Johnson, F. G. Lemoine, E. Mazarico, X. Sun, M. H. Torrence, A. M. Freed, C. Klimczak, J.-L. Margot, J. Oberst, M. E. Perry, R. L. McNutt, Jr., **§J. A. Balcerski**, **N. Michel**, M. J., Talpe, D. Yang, Topography of the northern hemisphere of Mercury from MESSENGER laser altimetry, *Science*, 336, 217-220, doi: 10.1126/science.1218805, 2012.
- Smith, D. E., M. T. Zuber, R. J. Phillips, S. C. Solomon, **S. A. Hauck, II**, F. G. Lemoine, E. Mazarico, G. A. Neumann, S. J. Peale, J.-L. Margot, C. L. Johnson, M. H. Torrence, M. E. Perry, D. D. Rowlands, S. Goossens, J. W. Head, A. H. Taylor, Gravity field and internal structure of Mercury from MESSENGER, *Science*, 336, 214-217, 2012.
- Peplowski, P. N., L. G. Evans, **S. A. Hauck, II**, T. J. McCoy, W. V. Boynton, J. J. Gillis-Davis, D. S. Ebel, J. O. Goldsten, D. K. Hamara, D. J. Lawrence, R. L. McNutt, L. R. Nittler, S. C. Solomon, E. A. Rhodes, A. L. Sprague, R. D. Starr, K. R. Stockstill-Cahill, Radioactive elements on Mercury's surface from MESSENGER: Implications for the planet's formation and evolution, *Science*, 333, 1850-1852, doi: 10.1126/science.1211576, 2011.
- **§Vilim, R., S. Stanley, S. A. Hauck, II**, Iron snow zones as a mechanism for generating Mercury's weak observed magnetic field, *Journal of Geophysical Research*, 115, E11003, doi: 10.1029/2009JE003528, 2010.
- Blewett, D. T., **S. A. Hauck, II**, H. Korth, Introduction to the special issue of *Icarus* on "Mercury after Two MESSENGER Flybys", *Icarus*, 209, 1-2, 2010.
- Zuber, M. T., L. G. J. Montesi, G. T. Farmer, **S. A. Hauck, II**, **§J. A. Ritzer**, R. J. Phillips, S. C. Solomon, D. E. Smith, M. J. Talpe, J. W. Head III, G. A. Neumann, T. R. Watters, C. L. Johnson, Accommodation of lithospheric shortening on Mercury from altimetric profiles of ridges and lobate scarps measured during MESSENGER flybys 1 and 2, *Icarus*, 209, 247-255-100, 2010.
- Smith, D. E., M. T. Zuber, R. J. Phillips, S. C. Solomon, G. A. Neumann, F. J. Lemoine, S. J. Peale, J. Margot, M. H. Torrence, M. J. Talpe, J. W. Head, **S. A. Hauck**, C. L. Johnson, M. E. Perry, O. S. Barnouin-Jha, R. L. McNutt, J. Oberst, The equatorial shape and gravity field of Mercury from MESSENGER flybys 1 and 2, *Icarus*, 209, 88-100, 2010.

- **§Ritzer, J. A., and S. A. Hauck, II**, Lithospheric structure and tectonics at Isidis Planitia, Mars, *Icarus*, 201, 528-539, 2009.
- Watters, T.R., S.C. Solomon, M.S. Robinson, J.W. Head, S.L. Andre, **S. A. Hauck, II**, and S.L. Murchie, The tectonics of Mercury: The view after MESSENGER's first flyby, *Earth and Planetary Science Letters*, 285, 283-296, 2009.
- Dombard, A. J., and **S. A. Hauck, II**, Despinning plus global contraction and the orientation of lobate scarps on Mercury, *Icarus*, 198, 274-276, 2008.
- Andrews-Hanna, J. C., M. T. Zuber, and **S. A. Hauck, II**, Strike-slip faults on Mars: Observations and implications for global tectonics and geodynamics, *Journal of Geophysical Research*, 113, E08002, doi: 10.1029/2007JE002980, 2008.
- Zuber, M. T., D. E. Smith, S. C. Solomon, R. J. Phillips, S. J. Peale, J. W. Head III, **S. A. Hauck, II**, R. L. McNutt, Jr., J. Oberst, G. A. Neumann, F. G. Lemoine, X. Sun, O. Barnouin-Jha, J. K. Harmon, Laser altimeter observations from MESSENGER's first Mercury flyby, *Science*, 321, 77-79, doi: 10.1126/science.1159086, 2008.
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- **Hauck, S. A., II**, S. C. Solomon, and **§D. A. Smith**, Predicted recovery of Mercury's internal structure by MESSENGER, *Geophysical Research Letters*, 34, L18201, doi: 10.1029/2007GL030793, 2007.
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- **Hauck, S. A., II**, J. M. Aurnou, and A. J. Dombard, Sulfur's impact on core evolution and magnetic field generation on Ganymede, *Journal of Geophysical Research*, 111, E09008, doi:10.1029/2005JE002557, 2006.
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